

ENTERPRISE AI KNOWLEDGE ASSISTANT

Enterprise Project Management Toolkit

Infrastructure & Capacity Plan

On-Premises / Private Deployment Sizing – EAIKA

Document Title	Infrastructure & Capacity Plan
Project Name	Enterprise AI Knowledge Assistant (EAIKA)
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1. Purpose

This document defines the infrastructure required to deploy and operate the Enterprise AI Knowledge Assistant (EAIKA) in production, including compute sizing for the local LLM (Qwen2.5 via Ollama), the ChromaDB vector store, and the Streamlit application tier, along with capacity planning assumptions for growth.

2. Deployment Topology

EAIKA is deployed entirely within the organization’s-controlled environment (on-premises data center or private cloud/VPC). No component requires outbound connectivity to a third-party AI API. The topology consists of three logical tiers: the application/UI tier (Streamlit + services layer), the LLM inference tier (Ollama hosting Qwen2.5), and the data tier (ChromaDB vector store and raw document storage).

3. Environment Overview

Environment	Purpose	Notes
Development	Feature development and unit testing.	Smaller Qwen2.5 model variant acceptable; sample document set only.
Test / QA	System, integration, and performance testing.	Mirrors production configuration; representative pilot document volume.
UAT / Staging	Business user acceptance testing.	Production-equivalent sizing; real (or de-identified) pilot documents.
Production	Live pilot deployment for PMO / project delivery teams.	Sized per capacity estimates below; monitored continuously.

4. Compute Sizing Guidance

Component	Sizing Consideration	Guidance
Ollama / Qwen2.5 LLM Host	Inference latency scales with model size and available GPU/CPU throughput.	GPU-accelerated host recommended for production to meet SLA response times; CPU-only acceptable for pilot-scale usage with relaxed latency expectations.
ChromaDB Vector Store	Index size grows with document volume and chunk count; query latency depends on memory availability.	Provision sufficient RAM to keep the active collection largely memory-resident; plan disk capacity for embeddings, metadata, and periodic snapshots.
Streamlit Application Tier	Handles concurrent user sessions and orchestrates the retrieval/generation flow.	Size for expected pilot concurrency with headroom for planned expansion to additional departments.
Document Storage (data/documents/)	Stores raw uploaded files pending and after processing.	Provision storage proportional to expected document volume growth; include standard backup retention.

5. Capacity Planning Assumptions

Assumption	Value / Basis
Pilot document volume at launch	Representative set of project management artifacts across active programs (charters, BRDs, status reports, RAID logs, UAT records).

Expected pilot concurrent users	Sized to the PMO / delivery team pilot population; expansion to additional departments planned for later phases.
Document growth rate	Incremental growth as new project documents are produced; periodic knowledge base rebuilds planned rather than continuous streaming ingestion.
Vector store growth	Proportional to chunk count, which scales with document volume and chosen chunk size/overlap settings.
Peak usage pattern	Expected around project status cycles, steering committee preparation, and UAT / release milestones.

6. Scaling Strategy

- Vertical scaling (additional CPU/GPU/RAM) on the LLM and vector store hosts is the primary lever for the pilot phase given the single-tenant, on-premises design.
- As document domains expand (HR, Finance, Healthcare, Operations), evaluate horizontal scaling of the application tier and a dedicated, higher-capacity vector store cluster.
- Knowledge base build/rebuild operations should be scheduled during low-usage windows to avoid contention with live query traffic.

7. Monitoring & Capacity Triggers

Metric	Monitoring Approach	Scale-Up Trigger
LLM response latency	Application-level timing on the generation step.	Sustained latency above the Enterprise Readiness & SLA target.
Vector store query latency	Application-level timing on the retrieval step.	Sustained latency degradation as document/chunk volume grows.
Host CPU / GPU / memory utilization	Infrastructure monitoring tooling.	Sustained utilization above 75–80% during business hours.
Storage utilization (documents + vector store)	Infrastructure monitoring tooling.	Utilization above 70% of provisioned capacity.

8. Backup & Disaster Recovery

- The ChromaDB persistent store and raw document repository are backed up on the organization's standard backup schedule and retention policy.
- Recovery procedures should include re-establishing the Ollama service and confirming the Qwen2.5 model is correctly loaded before restoring query traffic.
- A documented recovery time objective (RTO) and recovery point objective (RPO) should be agreed with Infrastructure/DevOps and recorded in the Enterprise Readiness & SLA document.

9. Approval

This Infrastructure & Capacity Plan requires sign-off from the Engineering Lead and Infrastructure/DevOps Lead prior to environment provisioning.